Basic Non-Sterile Compounding

for Pharmacists and Pharmacy Technicians: Day 1

Speaker: Lcdo. Jorge L. Reyes Quiñones, RPh

Institution: Santa Cruz Compounding Academy

Contact Hours: 6.0 | **Application Level:** 1 (Introductory)

Colegio de Farmacéuticos de Puerto Rico · Continuing Education Division

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ACPE Provider Number: 0151

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Provider Number	0151
Contact Hours (Day 1)	6.0 · Full course: 18 hours / 1.8 CEUs
Knowledge Level	1 (Introductory) — Application
Deadline to claim credits	45 days from the event
Validation	cfpr.org + CPE Monitor

General Rules and Regulations

ACPE-Accredited Continuing Education

1. Every participant must have registered their attendance.

2. Participants must remain physically present for the entire activity or until the speakers finish their presentation, since **no partial credits are awarded.**

3. The grace period for (2.0)-hour lectures is **(20) minutes.** On-site registration is not accepted after the corresponding grace period has ended.

4. Guests are not allowed in the lecture rooms.

5. The use of cell phones is not allowed in the lecture rooms. Please set them to silent mode; if you need to answer a call, please step out of the room to avoid disturbing your colleagues.

6. Recording the lecture is not allowed, neither video nor audio.

7. Remember that you have **(45) days** from the date of each lecture to claim your credits.

8. To claim your credits you must complete the Evaluation and the Test, which must be passed with **more than 70%.**

9. Log in at **www.cfpr.org**, find this lecture in the Course History and click the icon. First the Evaluation; once completed, the system authorizes taking the Test. After both steps are completed, the Certificate is released.

10. To claim credits with **ACPE / NABP**, access the **CPE Monitor** (through www.cfpr.org) and enter the **Access Code** provided at the end of the activity.

Instructional Methodology

Time distribution — Day 1 (6 contact hours)

- **Audiovisual lecture:** USP <795>, USP <800>, DQSA, Act 247, calculations.
- **Active learning:** pop questions, clinical cases, audience participation.
- **Hands-on laboratory:** hands-on compounding of 100 Progesterone SR capsules.

Block	Duration
Introduction + Rules + Disclosure	90 min
Module 1: Definitions, USP <795>/<800>, DQSA	90 min
Module 2: Act 247-2004 + Calculations	60 min
Laboratory: 100 Progesterone SR capsules	90 min
Post-test + Activity Evaluation Form	30 min

Objectives — Day 1

Upon completion of the CPE, the participant will be able to:

1. Define compounding and compare non-sterile compounding to sterile compounding.
2. Differentiate manufacturing compounding to pharmaceutical compounding.
3. Discuss in a detailed manner Chapters <795> and <800> of the United States Pharmacopeia.
4. Explain what Puerto Rico Pharmacy Law express about compounding in Puerto Rico.
5. Recognize the classification of compounding Pharmacies by the Drug Quality and Security Act (DQSA).
6. Review the calculation, formulation, procedures, and equipment used in the preparation of both, immediate and slow-release capsules.
7. Compound 100 slow-release capsules and check the weight deviations do not exceed the limits.

Needs Assessment

Needs analysis — why this course?

Why does it matter?	What to expect after the course?	What the participant should know	Application Activity
+67% growth in compounding Rx (2013–2023)	Confident application of USP <795>/<800> in daily practice	Definitions, BUDs, USP facilities, controls	100 Progesterone SR capsules + Pack-Stat Worksheet
Populations served: pediatric, geriatric, oncologic	Safe handling of hazardous drugs in the community pharmacy	NIOSH 2024 List, PPE, USP <800> engineering controls	HD clinical case (topical methotrexate) with PPE plan
Entrepreneurship opportunity in 503A compounding	Compliance with PR Act 247-2004 + federal DQSA	PR Board of Pharmacy, Regulation 156, 503A vs. 503B	Regulatory classification by clinical scenario
FDA products unavailable for customized doses	Mastery of pharmaceutical calculations for capsules	Pack stat, potency adjustment, fill weight, USP limits	Calculation set: adjusted API + USP verification

Source: NABP 2023 · ACPE Needs Assessment Guidelines · USP <795>/<800> · PR Act 247-2004

Activity Planning

Objectives · Practices · Final product as evidence

Objective	Practice / Experience 1	Practice / Experience 2	Final Product (Application Activity)
1. Define non-sterile vs sterile compounding	Classify 8 routes of administration	Identify dosage form in clinical cases	Documented categorized table
2. Differentiate mfg vs pharmaceutical compounding	Apply DQSA + NECC timeline	Classify scenario as 503A or 503B	Regulatory classification by Rx
3. Discuss USP <795> and <800>	Calculate BUD by dosage form	Identify NIOSH Table 1 HDs and required PPE	Documented MFR + PPE plan
4. Explain PR Act 247-2004	Identify PR Board of Pharmacy requirements	Regulation 156 checklist	Regulatory compliance sheet
5. Recognize 503A vs 503B	Side-by-side comparison	Solve DQSA pop questions	Classification by real-world scenarios
6. Review capsule calculations	Potency adjustment + DV	Pack stat scaling between sizes	Calculation set with USP verification
7. [LAB] Compound 100 Progesterone SR capsules	Determine pack stat (hand packing)	Verify % deviation (Pack-Stat Worksheet)	100 SR capsules + Pack-Stat Worksheet

Source: ACPE Standards 2015 · Bloom's Taxonomy (Application Level) · USP <795>/<800>

Competencies Assessed

Apply equally to pharmacists, technicians and students

PHARMACISTS · TECHNICIANS · STUDENTS — Same 6 competencies

1. Foundational knowledge
2. Population-based care
3. Promotion of wellness and health
4. Education
5. Innovation and Entrepreneurship
6. Professionalism

Post-test ≥ 70% required — uniform assessment for all profiles

Compounding: Definitions

Legal framework — DQSA Section 503A + Regulation 156, PR Dept. of Health

Pharmaceutical Compounding: combination, mixing or alteration of one or more ingredients by a pharmacist (or under their supervision) in response to a valid individual prescription, to meet the specific clinical need of an identified patient.

Non-Sterile (USP <795>)

- Oral: IR/SR capsules, suspensions, troches.
- Topical: creams, gels, ointments.
- Rectal / vaginal / otic / nasal.
- Sublingual / transdermal.

Sterile (USP <797>)

- IV: TPN, small-volume parenterals.
- IM: depot, sterile suspensions.
- Intrathecal / epidural.
- Ophthalmic / sterile nebulizations.

Source: USP <795> Nov 2023 · DQSA 2013 · PR Regulation 156

Compounding: Clinical Examples

Most common non-sterile routes of administration

Route	Dosage form	Clinical example
Oral	SR capsule	Topiramate 25 mg pediatric — refractory epilepsy
Oral	Flavored suspension	Omeprazole 2 mg/mL — reflux in an infant
Topical	Cream in Lipoderm®	Ketoprofen 10% — neuropathic pain
Rectal	Suppository	Diazepam 5 mg — febrile seizures
Vaginal	Suppository	Estriol 0.5 mg — postmenopausal vaginal atrophy
Sublingual	Troche	Sildenafil 50 mg — patient with dysphagia
Transdermal	Gel	Progesterone 100 mg in PLO — HRT

Source: Allen's Compounding Formulas · USP <795> Nov 2023

Manufacturing vs. Pharmaceutical Compounding

Regulatory framework under DQSA 2013

Feature	503A (Traditional)	503B (Outsourcing)
Regulatory body	PR Board of Pharmacy	FDA (cGMP 21 CFR 210/211)
Individual prescription	Yes — required	Not required
Scale	Per patient	Batches for hospitals/clinics
USP standards	<795> / <797> / <800>	<795> / <797> / <800> + cGMP
FDA inspection	Episodic	Routine (inspection program)
Interstate distribution	≤ 5% of prescriptions	No percentage limit
Copying FDA-approved products	Prohibited (shortage exception)	Permitted under FDA list

Source: DQSA 2013 H.R. 3204 · FDA · NCBI-NIH StatPearls 2020

USP <795> — Effective Nov 1, 2023

Pharmaceutical Compounding — Nonsterile Preparations

Facilities

- Designated area separate from dispensing.
- Temperature ≤ 25°C, humidity 30–60%.
- Non-porous, cleanable surfaces.

Personnel — Designated Person (DP)

- Designated Person mandatory (new 2023 role).
- Documented training.
- Annual competency assessment.

Components / APIs

- CoA required for each API.
- USP-NF identity testing.
- Do not copy FDA-approved products.

Documentation

- Master Formulation Record (MFR).
- Compounding Record per batch.
- Adverse events + equipment logs.

Source: USP <795> Nov 2023 · PCCA 2023 · NABP

USP <795> — Beyond-Use Dates (BUDs)

Changes effective since November 2023

Preparation type	Aw (water activity)	Default BUD
Aqueous oral without preservative	Aw ≥ 0.6	14 days
Preserved aqueous oral	Aw ≥ 0.6	35 days
Nonaqueous oral	Aw < 0.6	90 days
Nonaqueous: capsules, topicals, suppositories	Aw < 0.6	180 days
With validated stability data	—	Extendable (per data)

Clinical note: If stability data from an FDA-registered laboratory is available, the BUD may be extended beyond the default value. Document the reference and keep it in the MFR.

Source: USP <795> Nov 2023 · PCCA 2023 · Compounding Center Gill & Garvin Oct 2023

QUICK QUESTION #1 · Learning Check

USP <795> — Beyond-Use Dates

According to USP <795> (November 2023), what is the maximum default BUD for an aqueous oral preparation WITHOUT preservative and without specific stability information?

A) 30 days

B) 14 days

C) 180 days

D) 35 days

ANSWER — QUICK QUESTION #1

USP <795> — Beyond-Use Dates

According to USP <795> (November 2023), what is the maximum default BUD for an aqueous oral preparation WITHOUT preservative and without specific stability information?

A) 30 days

B) **14 days ✓ CORRECT**

C) 180 days

D) 35 days

Rationale: USP <795> 2023 sets a maximum BUD of 14 days for aqueous oral preparations without preservative (Aw ≥ 0.6).

USP <800> — Hazardous Drugs

NIOSH 2024 classification (CDC/NIOSH Pub 2025-103)

TABLE 1 — Antineoplastics

- IARC Group 1 / 2A.
- **Handling ALWAYS requires ALL USP <800> controls.**
- Examples: cyclophosphamide, methotrexate, doxorubicin.
- Maximum PPE + C-PEC in C-SEC with negative pressure.

TABLE 2 — Other FDA-CDER HDs

- Other drugs with identified risk.
- **Eligible for Assessment of Risk (AOR).**
- AOR reviewable ≥ 12 months.
- Document: HD type, dosage form, exposure risk, packaging, handling.

Source: USP <800> · NIOSH 2024 (CDC/NIOSH Pub 2025-103) · Wolters Kluwer 2025

USP <800> — PPE and Engineering Controls

Mandatory equipment for handling HDs

Mandatory PPE

- Double gloves, ASTM D6978.
- Laminated polyethylene gown, non-reusable.
- Head / hair / shoe / sleeve covers.
- Certified goggles.
- Surgical N95 respirator (minimum).

Engineering controls

- C-PEC inside the C-SEC.
- Negative pressure: -0.01 to -0.03 in H_2O.
- Minimum 12 ACPH (air changes per hour).
- External HEPA exhaust (not recirculated).
- CSTD for antineoplastics during administration.

Spill management (Spill Kit + SOP): 1) Isolate the area · 2) IPA · 3) H_2O · 4) H_2O_2 · 5) EPA-registered disinfectant · Document the incident.

Source: USP <800> · NIOSH 2024 · OSHA HD Guidelines · Duke Safety 2025

Clinical Case — Hazardous Drug Handling

Applying USP <800> in 503A practice

Patient description

58-year-old adult male diagnosed with severe psoriatic arthritis refractory to biologic therapy. The rheumatologist prescribes: Methotrexate 0.5% topical cream in Lipoderm® base — 30 g, apply BID. No FDA-approved commercial topical presentation exists → 503A compounding indicated.

USP <800> compounding plan

Classification: Methotrexate — NIOSH Table 1 (antineoplastic, high risk).
PPE: double gloves ASTM D6978, impermeable gown, N95, goggles, head/shoe covers.
Controls: C-PEC in C-SEC with negative pressure, minimum 12 ACPH, external HEPA exhaust.
Packaging: opaque container with child-resistant seal; CYTOTOXIC label.
Counseling: wear gloves when applying, contraindicated in pregnancy, discard as HD, wash hands.

Source: USP <800> · NIOSH 2024 Table 1 · OSHA HD Guidelines · Allen's

PR Pharmacy Act No. 247-2004

Local legal framework for compounding in Puerto Rico

Legal definition of compounding: Regulation No. 156, PR Dept. of Health (2022).

Regulatory authority: Puerto Rico Board of Pharmacy (registration, supervision, inspections, penalties).

Registration requirements: active pharmacy license, identified supervising pharmacist, registered technicians, compliance with USP standards.

Mandatory reporting: adverse events to the Dept. of Health, recalls, violations, periodic inspections.

Penalties: license suspension or revocation, administrative fines, civil and criminal liability, closure of the pharmacy.

Integrated compliance: Act 247 (PR) + DQSA (federal) + USP <795>/<800> (technical standard)

Source: Act No. 247-2004 · Regulation 156 · PR Dept. of Health · cfpr.org/documents.cfm

DQSA — Section 503A vs. 503B

Expanded comparison table

Criterion	503A — Traditional Compounding	503B — Outsourcing Facility
Primary regulator	State Board of Pharmacy	FDA (cGMP 21 CFR 210/211)
Individual prescription	Yes — required (identified patient)	No — anticipatory batches
FDA inspections	Episodic	Routine (inspection program)
Interstate distribution	≤ 5% of volume (without MOU)	No percentage limit
Copying FDA-approved products	Prohibited (shortage exception)	Permitted under FDA list
USP compliance	<795>, <797>, <800>	<795>, <797>, <800> + cGMP
Sterile compounding	Permitted under USP <797>	Permitted + cGMP

Source: DQSA 2013 H.R. 3204 · FDA · NCBI-NIH StatPearls 2020 · Fagron Sterile 2024

DQSA — Regulatory Timeline

Key milestones in compounding regulation

1997	FDAMA Section 503A — first attempt at a federal framework.
2002	Thompson v. Western States — Supreme Court declares part of 503A unconstitutional.
Sep 2012	NECC outbreak — 64 deaths from fungal meningitis.
Nov 2013	DQSA signed — creates 503B and strengthens 503A.
Nov 2023	USP <795> revised — new BUDs, Designated Person, documented AOR.
Dec 2024	NIOSH 2024 List of HDs — updated (CDC/NIOSH Pub 2025-103).

Source: FDA · CDC · USP · NIOSH 2024 · NCBI-NIH

QUICK QUESTION #2 · Learning Check

DQSA Classification — 503A vs. 503B

A community pharmacy in PR compounds bioidentical hormone capsules under an individual prescription. Which DQSA classification applies and who primarily regulates it?

A) 503B — regulated by FDA cGMP

B) 503A — regulated by the PR Board of Pharmacy

C) 503A — regulated exclusively by FDA

D) 503B — no inspection required

ANSWER — QUICK QUESTION #2

DQSA Classification — 503A vs. 503B

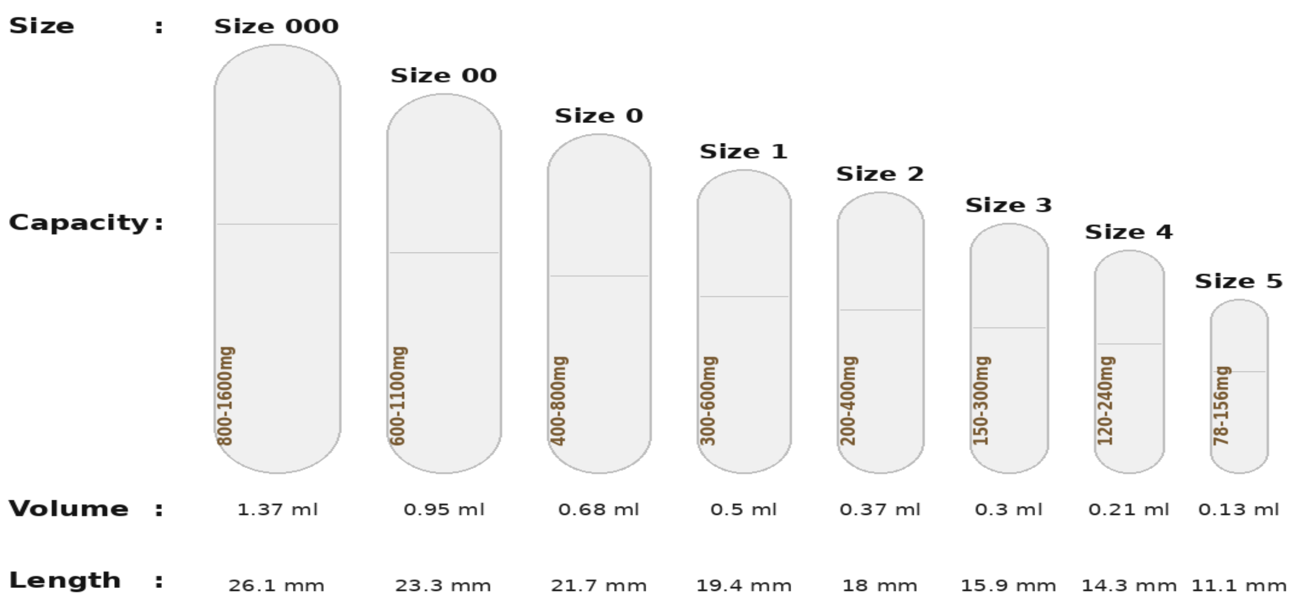
A community pharmacy in PR compounds bioidentical hormone capsules under an individual prescription. Which DQSA classification applies and who primarily regulates it?

A)	503B — regulated by FDA cGMP
B)	**503A — regulated by the PR Board of Pharmacy ✓ CORRECT**
C)	503A — regulated exclusively by FDA
D)	503B — no inspection required

Rationale: A community pharmacy with an individual Rx is 503A under DQSA. The primary regulator is the state (PR) Board of Pharmacy.

Capsule Sizes

Capacity, volume and length of capsules 000 to 5



Size 000 (largest, 800–1600 mg) through size 5 (smallest, 78–156 mg). Selection depends on the total weight of the formula + the pack stat of the API.

Source: PCCA Capsule Sizing Reference · Allen's Compounding Formulas

Conversion Factors — Pack Stats

Multiply the #1 capsule pack stat by the factor to scale to other sizes

Rule of thumb: *Pack stat (size X) = Pack stat (capsule #1) × Conversion factor*

Capsule Size	Conversion Factor
Capsule 1 → Capsule 0	**1.37**
Capsule 1 → Capsule 00	**1.95**
Capsule 1 → Capsule 000	**2.95**
Capsule 1 → Capsule 3	**0.61**

Example: Progesterone pack stat in #1 = 270 mg → in #0 = 270 × 1.37 = 369.9 mg · in #00 = 270 × 1.95 = 526.5 mg

Source: Allen's Compounding Formulas · PCCA Capsule Sizing Reference · USP <795>

Capsule Calculations (1/2)

API Potency and Displacement Value (DV)

API Potency Adjustment

Formula: API quantity = Dose × (100 ÷ Potency%)

Example: Dose 50 mg, CoA potency 97.5% → 50 × (100 / 97.5) = 51.28 mg/capsule

Displacement Value (DV)

Formula: Diluent vol. = Fill Weight − (API qty ÷ DV)

Example: API 100 mg, DV 1.5, Fill Weight 400 mg → Diluent = 400 − (100 ÷ 1.5) = 333 mg

The DV is obtained from standard tables (Allen's, PCCA) or measured experimentally when it is not reported.

Source: USP <795> 2023 · Allen's Compounding Formulas · PCCA

Capsule Calculations (2/2)

Fill Weight by size + USP deviation limits

Capsule size	Typical fill weight (mg)
#000	950 – 1100
#00	700 – 900
#0	400 – 600
#1	300 – 400
#2	200 – 300

USP Limits — Weight Deviation

Fill ≥ 300 mg:

Tolerance ± 7.5%

Fill < 300 mg:

Tolerance ± 10%

20-capsule rule:

NMT 2 capsules outside the limit.

No capsule at twice the limit.

Source: USP <795> 2023 · USP <1> Weight Variation · Allen's

QUICK QUESTION #3 · Learning Check

API Calculation + USP Lower Limit

Patient: 45-year-old adult female. Rx: SR capsules 300 mg API · CoA potency = 96% · Theoretical fill weight = 500 mg.

How much API is weighed per capsule and what is the USP lower limit for total weight?

A) API: 300 mg | Lower limit: 455 mg

B) API: 312.5 mg | Lower limit: 462.5 mg

C) API: 312.5 mg | Lower limit: 450 mg

D) API: 306 mg | Lower limit: 470 mg

Solution: API = 300 × (100/96) = 312.5 mg. Fill ≥ 300 mg → ±7.5%. 500 × 0.925 = 462.5 mg (lower limit).

ANSWER — QUICK QUESTION #3

API Calculation + USP Lower Limit

How much API is weighed per capsule and what is the USP lower limit for total weight?

A)	API: 300 mg | Lower limit: 455 mg
B)	**API: 312.5 mg | Lower limit: 462.5 mg ✓ CORRECT**
C)	API: 312.5 mg | Lower limit: 450 mg
D)	API: 306 mg | Lower limit: 470 mg

Laboratory Equipment (1/2)

Analytical Balance

Analytical Balance

Class: Class I / II NIST / USP — precision ± 0.1 mg.

Calibration: daily, documented, with certified NIST weights.

Mandatory logbook: date, operator, value obtained vs. expected, corrective action.

Location: stable surface, free of air currents, away from vibrations.

Pre-use verification: leveling, tare, zero-point calibration.

Documentation: NIST traceability, weight certificates, annual preventive maintenance.

Source: USP <795> 2023 · PCCA Equipment Guide · Allen's

Laboratory Equipment (2/2)

V-Blender Pro Mixer and Profiler Capsule Filler

MAZ Mixer

- Homogeneous powder mixing.
- Deaeration or elimination of bubbles.
- Speed: up to 400-2200 rpm.
- Time: 60-90 seconds
- Speeds up the mixing process for filling capsules.

Profiler Capsule Filler 100/300

- Capacity: 100 or 300 capsules / batch.
- Sizes: 000 – 4.
- Consistent fill per capsule.
- Easy cleaning between sizes.
- Calibrate the theoretical weight before the batch.
- Verify complete closure of each capsule.

Source: USP <795> 2023 · PCCA Equipment Guide

Procedure — 100 Slow-Release Capsules (1/2)

Steps 1 to 4: preparation and weighing

1. **Verify documents**

   MFR, Rx, CoA, API BUD, theoretical fill weight, capsule size.

2. **Prepare area and PPE**

   Clean the surface, gloves, gown, calibrate the balance with certified NIST weights.

3. **Weigh potency-adjusted API**

   Quantity = Dose × (100/Potency%) × 100. Document; double-check by a 2nd operator.

4. **Weigh SR excipient**

   Calculate using DV and theoretical fill weight. Example: HPMC K4M, MCC PH-101, colloidal silica.

Source: USP <795> 2023 · PCCA SOP · Allen's Compounding

Procedure — 100 Slow-Release Capsules (2/2)

Steps 5 to 8: mixing, filling and verification

5 **Mix in the MAZ**

Mix 3 times and verify visual homogeneity

6 **Fill in the Profiler 100**

Uniform distribution; correct tamping; verify complete closure of each capsule.

7 **Verify USP weights**

Weigh 10 capsules. Calculate % deviation. Apply ±7.5% (≥300 mg) or ±10% (<300 mg). PASS / FAIL.

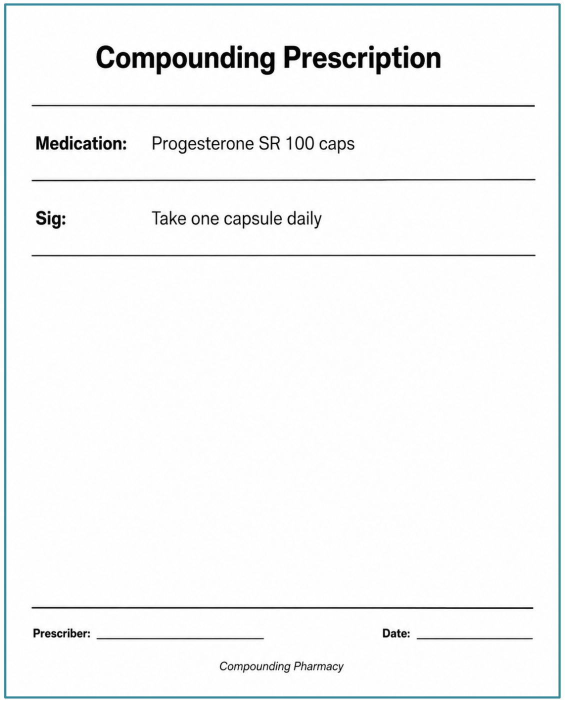
8 **Label and document**

Act 247 Rx label (patient, BUD, storage). Complete Compounding Record. File for ≥ 1 year.

Source: USP <795> 2023 · USP <1> · PCCA SOP

Application Activity — Laboratory

Practice formula: Progesterone SR 100 capsules



Compounding Prescription

Medication: Progesterone SR 100 caps

Sig: Take one capsule daily

Prescriber: ____________ **Date:** ________

Compounding Pharmacy

- **Sig:** take one capsule daily — Progesterone SR 100 capsules.
- **Pack stat:** determine by hand packing (Hand Pack Procedure, Letco Medical).
- **Worksheet:** complete the Pack-Stat Determination Worksheet — Methocel, MCC and Lactose (Progesterone).
- **USP verification:** % weight deviation across 10 capsules per USP acceptance criteria.
- **Final product:** 100 SR capsules + Compounding Record + Pack-Stat Worksheet + Act 247 label.

Source: Letco Medical — Capsule Packing Statistics Procedure · Pack-Stat Determination Worksheet (Santa Cruz Compounding Academy)

Reference — Capsule Packing Statistics Procedure

Letco Medical — Hand-packing determination method

Capsule Packing Statistics Procedure

Compounders are frequently asked to formulate a capsule dosage form using a manufactured tablet or capsule as the starting active pharmaceutical ingredient (API) source or a bulk API with unknown capsule packing statistics. In order to develop a proper capsule formulation the capsule packing statistics for any unknown ingredients must first be determined. Capsule packing statistics should also be determined when receiving new lots of chemical from a chemical vendor. This is suggested for each new lot number as variances do occur from lot to lot even from the same supplier.

Capsule packing statistics for a powder, whether from crushed tablets, capsule contents, bulk API's or excipients can be determined by hand packing empty capsules. Letco recommends using a #1 capsule for the process as the results can be extrapolated to determine capsule packing statistics for other size capsules.

Hand Pack Procedure	
1	Determine average empty weight of capsule (top and bottom). Weigh 10 empty capsules then divide the total weight by 10.
2	Prepare powder by placing a sufficient amount in to a medium size weigh boat. Use a spatula to compress powder to approximately ¼ inch in height. (If material is crystalline or granular form, reduce particle size in a mortar before packing)
3	Firmly press capsule bottom into powder with a twisting motion. Repeat until capsule is completely full.
4	Recap capsule and repeat the process until 5 packed capsules are obtained.
5	Weigh each capsule and record weights.
6	Subtract average empty capsule weight from each recorded weight in Step #5.
7	Determine average weight (Sum of 5 weights/ 5)

To extrapolate the #1 capsule packing statistic to a different size capsule, multiply the #1 capsule packing statistics by the appropriate factor indicated in the table below.

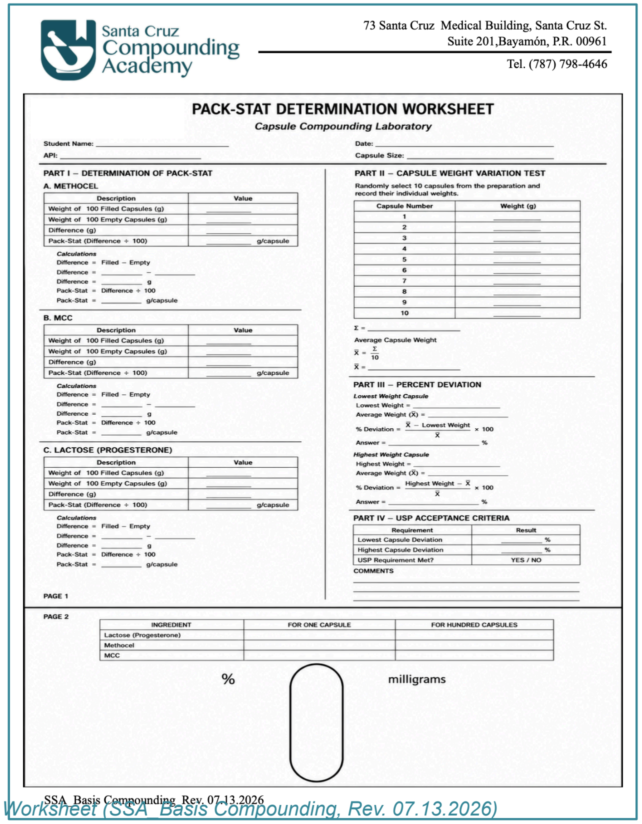
Capsule Size	#5	#4	#3	#2	#1	#0	#00	#000
Factor	0.27	0.43	0.61	0.74	------	1.37	1.95	2.95

Factor calculations are based on known capsule physical volume with slight adjustments to account for differences in packing from person to person. Results are approximations that may vary slightly from the true statistic due to a variety of factors.

Source: Letco Medical Compounding Solutions — Capsule Packing Statistics Procedure

Appendix — Pack-Stat Determination Worksheet

Laboratory worksheet — Santa Cruz Compounding Academy



Santa Cruz Compounding Academy

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Suite 201,Bayamón, P.R. 00961
Tel. (787) 798-4646

PACK-STAT DETERMINATION WORKSHEET

Capsule Compounding Laboratory

Student Name: ____________________ Date: ____________________

API: ____________________ Capsule Size: ____________________

PART I – DETERMINATION OF PACK-STAT

A. METHOCEL

Description	Value
Weight of 100 Filled Capsules (g)	________
Weight of 100 Empty Capsules (g)	________
Difference (g)	________
Pack-Stat (Difference ÷ 100)	________ g/capsule

Calculations

Difference = Filled – Empty
Difference = ________ – ________
Difference = ________ g
Pack-Stat = Difference ÷ 100
Pack-Stat = ________ g/capsule

B. MCC

Description	Value
Weight of 100 Filled Capsules (g)	________
Weight of 100 Empty Capsules (g)	________
Difference (g)	________
Pack-Stat (Difference ÷ 100)	________ g/capsule

Calculations

Difference = Filled – Empty
Difference = ________ – ________
Difference = ________ g
Pack-Stat = Difference ÷ 100
Pack-Stat = ________ g/capsule

C. LACTOSE (PROGESTERONE)

Description	Value
Weight of 100 Filled Capsules (g)	________
Weight of 100 Empty Capsules (g)	________
Difference (g)	________
Pack-Stat (Difference ÷ 100)	________ g/capsule

Calculations

Difference = Filled – Empty
Difference = ________ – ________
Difference = ________ g
Pack-Stat = Difference ÷ 100
Pack-Stat = ________ g/capsule

PAGE 1

PART II – CAPSULE WEIGHT VARIATION TEST

Randomly select 10 capsules from the preparation and record their individual weights.

Capsule Number	Weight (g)
1	________
2	________
3	________
4	________
5	________
6	________
7	________
8	________
9	________
10	________

Σ = ____________

Average Capsule Weight

$\bar{X} = \frac{\Sigma}{10}$

$\bar{X}$ = ____________

PART III – PERCENT DEVIATION

Lowest Weight Capsule

Lowest Weight = ____________
Average Weight ($\bar{X}$) = ____________

$\% \text{ Deviation} = \frac{\bar{X} - \text{Lowest Weight}}{\bar{X}} \times 100$

Answer = ____________ %

Highest Weight Capsule

Highest Weight = ____________
Average Weight ($\bar{X}$) = ____________

$\% \text{ Deviation} = \frac{\text{Highest Weight} - \bar{X}}{\bar{X}} \times 100$

Answer = ____________ %

PART IV – USP ACCEPTANCE CRITERIA

Requirement	Result
Lowest Capsule Deviation	________ %
Highest Capsule Deviation	________ %
USP Requirement Met?	YES / NO

COMMENTS

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INGREDIENT	FOR ONE CAPSULE	FOR HUNDRED CAPSULES
Lactose (Progesterone)		
Methocel		
MCC		

% milligrams

SSA_Basis Compounding_Rev. 07.13.2026

Source: Santa Cruz Compounding Academy — Pack-Stat Determination Worksheet (SSA_Basis Compounding, Rev. 07.13.2026)

Assessment

Learning assessment methods (ACPE)

- **Pre-test at the start** — knowledge baseline.
- **Pop questions during the session** — 3 interspersed (USP, DQSA, calculation).
- **Case discussion + active participation** — cases with patient description.
- **Laboratory activity report** — Pack-Stat Worksheet + weight deviation report.
- **Post-test at the end** — ≥ 70% passing score to earn ACPE credits.
- **Activity Evaluation Form (CFPR)** — required by ACPE.

≥ 70% passing score on the post-test · 45 days to claim credits at cfpr.org + CPE Monitor

Summary — Day 1

Six key points of the module

1. Compounding: definitions + 503A vs. 503B (DQSA).
2. USP <795> 2023: updated BUDs (14/35/90/180 days).
3. USP <800>: HD handling + PPE + engineering controls.
4. PR Act 247-2004 + DQSA: complete legal framework.
5. Capsule calculations: potency, DV, fill weight, USP limits.
6. CPE reminder: post-test ≥ 70%, 45 days, cfpr.org.

Packing Statistics

Key concept for capsule formulation

Packing statistic (pack stat): the weight of a specific ingredient that occupies the total volume of a capsule of a given size, under standard packing conditions.

Where does the pack stat come from?

- **API supplier / distributor:** packing statistics table supplied with the lot (CoA or technical data sheet).
- **Calculated in the laboratory:** weigh the API that fills 10 empty capsules using standard hand packing; average the result.
- It does NOT appear in a USP monograph, nor is it the responsibility of USP/FDA.

Source: USP <795> 2023 · Allen's Compounding Formulas · PCCA Capsule Guide

Packing Statistics — Example (1/2)

% of capsule volume occupied by the API

Clinical question

If the pack stat of progesterone is 270 mg in a size #1 capsule, what % of the capsule volume do 100 mg of progesterone occupy?

Solution

Rule: % volume = (API weight / pack stat) × 100

Calculation: (100 mg / 270 mg) × 100 = 37.0%

Result: 100 mg of progesterone occupy ~37% of the volume of a #1 capsule. The remaining 63% is filled with diluent (Methocel, MCC, silica, etc.).

Source: USP <795> 2023 · Allen's Compounding Formulas

Packing Statistics — Example (2/2)

Scaling the pack stat between capsule sizes

Clinical question

Progesterone pack stat = 270 mg in a #1 capsule. The volume of a #0 capsule is 1.95× the volume of a #1. What is the pack stat for a #0 capsule?

Solution

Proportion rule: pack stat #0 = pack stat #1 × volume ratio

Calculation: 270 mg × 1.95 = 526.5 mg

Result: The pack stat of progesterone in a #0 capsule is 526.5 mg. The proportion applies to ANY ingredient across capsule sizes.

Source: USP <795> 2023 · Allen's · PCCA Capsule Sizing

Methocel · Hypromellose · Slow release

Key excipient in extended-release capsules

Methocel® = Hypromellose = HPMC (Hydroxypropyl Methylcellulose). Hydrophilic polymer that forms a gel matrix upon hydration → releases the API slowly.

Key properties

- **It is the ingredient RESPONSIBLE for extended release in SR capsules.**
- Grades by viscosity: K100LV, K4M, K15M, K100M (the higher the M, the higher the viscosity and the greater the SR).
- K4M and K15M are the most used for 8–12 h SR.
- Typical concentration in SR capsules: 10–30% of total weight.
- Mechanism: hydration → gel → controlled diffusion of the API.

Source: USP <795> 2023 · Dow Chemical Methocel Technical Handbook · Allen's

USP <800> — Full Scope + Act 247

Regulatory details for the post-test

USP <800> applies to ALL phases

- Storage of HDs.
- Compounding (sterile and non-sterile).
- Administration (including IV).
- Disposal / discarding of HD preparations.
- Transport, spills, transfer.

PR Act 247-2004 — Compounding

- **Compounding in anticipation of a prescription is NOT allowed.**
- Each preparation requires an individual Rx and an identified patient.
- No stock for general sale — only in response to a specific Rx.
- Quantity limit: only what is needed per patient and therapeutic duration.
- Mandatory registration with the PR Board of Pharmacy + applicable USP <795>.

Source: USP <800> · Act No. 247-2004 · Regulation 156 · PR Board of Pharmacy

POST-TEST

Day 1 · 15 questions
Select the BEST answer
Passing score required: ≥ 70%
One (1) question per slide

POST-TEST · Question 1 / 15

USP Chapter — Non-Sterile Compounding

This USP chapter describes the minimum standards to follow for the preparation of non-sterile compounded formulations (human and animal):

A) 513

B) 795

C) 797

D) 800

E) none of the above is correct

POST-TEST · Question 2 / 15

USP Chapter — Hazardous Drugs

This chapter describes practice and quality standards for handling hazardous drugs, to promote patient safety, worker safety and environmental protection:

- **A)** 513
- **B)** 795
- **C)** 797
- **D)** 800
- **E)** none of the above

POST-TEST · Question 3 / 15

List of Hazardous Drugs

The list of hazardous drugs is prepared by:

- **A)** USP
- **B)** FDA
- **C)** EPA
- **D)** NIOSH
- **E)** OSHA

Required PPE

For compounding sterile and non-sterile hazardous drugs, the required PPE is:

A) gowns

B) head and hair covers

C) shoe covers

D) two pairs of chemotherapy gloves

E) all of the above are correct

POST-TEST · Question 5 / 15

PR Pharmacy Act

Under the Puerto Rico Pharmacy Act, with respect to compounding, the law allows you to:

A)	prepare only a limited quantity of a compounded preparation for inventory
B)	you may not compound in anticipation of a prescription
C)	the number of units to prepare depends on the BUD of the preparation
D)	the number of units depends on the frequency of prescriptions per day
E)	none of the above is correct

POST-TEST · Question 6 / 15

DQSA

DQSA:

- **A)** refers to the Drug Quality Security Act
- **B)** significantly strengthened FDA authority over pharmaceutical compounding
- **C)** creates a new "Outsourcing Facility" category
- **D)** was enacted in 2013
- **E)** all of the above are correct

POST-TEST · Question 7 / 15

Capsule Formulation Knowledge

For capsule formulation we must know:

- **A)** the density of the API
- **B)** the density of all the ingredients in the capsule
- **C)** the packing statistic of all the ingredients in the capsule
- **D)** the hazardous ingredients in the capsule
- **E)** none of the above is correct

POST-TEST · Question 8 / 15

Capsule BUD

The BUD assigned to capsules is usually:

A)	30 days
B)	90 days
C)	180 days or the expiration date of any ingredient if it is less than 180 days (whichever is shorter)
D)	180 days
E)	none of the above is correct

POST-TEST · Question 9 / 15

Source of the Packing Statistic

The packing statistic of an ingredient:

- **A)** may be provided by the distributor
- **B)** may be calculated in the laboratory
- **C)** is part of the ingredient's monograph
- **D)** is provided by USP
- **E)** A and B are correct

Pack Stat — % of Volume

If the pack stat of progesterone is 270 mg for a size 1 capsule, what percentage of the capsule volume will 100 mg of progesterone occupy?

- **A)** 3.7%
- **B)** 37%
- **C)** 370%
- **D)** 63%
- **E)** none of the above is correct

POST-TEST · Question 11 / 15

Pack Stat — Scaling

If the pack stat of progesterone is 270 mg for a size 1 capsule and the volume of a size 0 capsule is 1.95 times the volume of size 1, what will the pack stat of the size 0 capsule be?

A) 271.95 mg

B) 268.05 mg

C) 526.5 mg

D) 138.46 mg

E) none of the above is correct

POST-TEST · Question 12 / 15

Methocel Identity

Methocel is equivalent to Hypromellose.

A) True

B) False

POST-TEST · Question 13 / 15

Methocel Mechanism

Methocel is the ingredient responsible for slow-release capsules.

A)	True
B)	False

POST-TEST · Question 14 / 15

Compounding and Drug Shortages

Compounding pharmacies are not allowed to make commercially available products. If a commercially available product is in shortage, the compounding pharmacy may duplicate the product until the shortage ends.

A) True

B) False

POST-TEST · Question 15 / 15

503A / 503B — Classification

Compounding pharmacies are classified as 503A and 503B: 503A compounds sterile products; 503B compounds non-sterile products.

A)	True
B)	False

ANSWER KEY — POST-TEST (1 of 2)

For facilitator use / participant self-assessment · Questions 1–8

Q#	Answer	Rationale
Q1	**B**	USP General Chapter <795> sets the minimum standards for non-sterile compounded preparations (human and veterinary); <797> covers sterile compounding and <800> covers hazardous drugs.
Q2	**D**	USP <800> establishes the practice and quality standards for handling hazardous drugs to protect patients, personnel, and the environment.
Q3	**D**	NIOSH publishes and periodically updates the List of Hazardous Drugs in Healthcare Settings (2024 list, CDC/NIOSH Pub 2025-103), which USP <800> incorporates by reference.
Q4	**E**	USP <800> requires full PPE for hazardous-drug compounding — gowns, head/hair covers, shoe covers, and two pairs of ASTM D6978 chemotherapy gloves — for both sterile and non-sterile preparations.
Q5	**B**	PR Act 247-2004 prohibits compounding a preparation in anticipation of a prescription; every preparation must be tied to an individual Rx and an identified patient.
Q6	**E**	DQSA (Drug Quality and Security Act, enacted Nov. 2013) strengthened FDA authority over compounding and created the new 503B outsourcing-facility category — B, C, and D are all correct.
Q7	**C**	Formulating a capsule requires the packing statistic of every ingredient, since that determines how much of each fills the capsule shell — not density or hazard classification.
Q8	**C**	USP <795> assigns capsules a default BUD of 180 days, but that default is capped by the expiration date of any component ingredient if it falls sooner than 180 days.

ANSWER KEY — POST-TEST (2 of 2)

For facilitator use / participant self-assessment · Questions 9–15

Q#	Answer	Rationale
Q9	**E**	The pack stat is either supplied by the API distributor (CoA/technical data sheet, A) or determined experimentally in the lab by hand-packing (B) — it is not published in a USP monograph.
Q10	**B**	% volume = (API weight ÷ pack stat) × 100 = (100 mg ÷ 270 mg) × 100 = 37%.
Q11	**C**	Pack stat scales proportionally with capsule volume: 270 mg × 1.95 (the #0-to-#1 volume ratio) = 526.5 mg.
Q12	**A**	Methocel® is the trade name for hypromellose (HPMC) — the two terms refer to the same polymer.
Q13	**A**	Methocel (HPMC) hydrates into a gel matrix that controls and slows drug release, making it the key excipient responsible for the SR effect.
Q14	**A**	Compounding a copy of an FDA-approved product is normally prohibited, but the FDA Drug Shortage List creates a documented exception that allows it for as long as the shortage lasts.
Q15	**B**	This is a common myth — 503A vs. 503B is not a sterile-vs.-non-sterile split. Both can compound either type; the real difference is the regulatory model (individual Rx + state Board of Pharmacy vs. anticipatory batches + FDA cGMP).

Each question answered correctly represents demonstration of achievement of the corresponding objective.

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How to Obtain the Certificate

Validation at cfpr.org + CPE Monitor

1. Complete the post-test at cfpr.org (≥ 70% to pass).
2. Complete the Activity Evaluation Form.
3. Log in at cfpr.org → MY ACCOUNT → COURSE HISTORY.
4. Select the course and enter the access code.
5. CPE Monitor syncs credits automatically with NABP.
6. Save or print the certificate from CPE Monitor.

DEADLINE: 45 days after the event, No exceptions after this deadline.

Thank you for your participation!

Day 1 completed — Basic Non-Sterile Compounding

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